

Sovesh Mohapatra

Department of Bioengineering
University of Pennsylvania
soveshm@engineering.upenn.edu
+1 (857) 389-8786 · soveshmohapatra.com
GitHub · Google Scholar · LinkedIn

EDUCATION

Ph.D. Bioengineering
University of Pennsylvania, May 2027 (expected)
Advisor: Dani S. Bassett

B.S. Computer Science & Mathematics
B.A. Linguistics
University of Massachusetts Amherst, 2023

RESEARCH INTERESTS

I develop computational methods to study how network structure shapes dynamics and function in the brain. My research combines network control, machine learning for neuroimaging, and large-scale analysis of scientific communication.

PUBLICATIONS

Sovesh Mohapatra and Dani S. Bassett. **Task Context Shapes Component Roles in Language Models.** *Under review at the International Conference on Learning Representations (ICLR), 2027.*

Sovesh Mohapatra and Dani S. Bassett. **Connectome as Compiler: Reading Computation from Connectivity and then Writing it Back.** *Preprint.* 2026.

Sovesh Mohapatra, David Lydon-Staley, and Dani S. Bassett. **War in the Abstract: The rise and consequences of militarized language in scientific communication.** *Under consideration at PNAS.* 2026.

Sovesh Mohapatra, Christoffer G. Alexandersen, Panagiotis Fotiadis, Max B. Kelz, John A. Detre, Fabio Pasqualetti, and Dani S. Bassett. **QUIET: Quantifying Underutilized Influential Edges for Targeted synchronization.** *Under consideration at Nature Computational Science.* 2026.

Sovesh Mohapatra, Isabella Llano Aristizabal, Christoffer G. Alexandersen, John A. Detre, and Dani S. Bassett. **Edge-centric control energy follows an inverted-U trajectory through Alzheimer's disease progression.** *Under consideration at Entropy.* 2026.

Sovesh Mohapatra, Advait Gosai, Anant Shinde, Aleksei Rutkovskii, Sirisha Nouduri, and Gottfried Schlaug. **Meta-analysis of transfer learning for segmentation of brain lesions.** *Under consideration at PLOS Computational Biology.* 2026.

Sovesh Mohapatra, Minhui Ouyang, Shufang Tan, Jianlin Guo, Lianglong Sun, Yong He, and Hao Huang. **TReND: Transformer-derived features and Regularized NMF for neonatal functional network delineation.** *MICCAI.* 2025.

Rina Bao, Anna N. Foster, Ya'Nan Song, Rutvi Vyas, Ankush Kesri, Imad Eddine Toubal, Elham Soltani Kazemi, Gani Rahmon, Taci Kucukpinar, Mohamed Almansour, Mai-Lan Ho, K. Palaniappan, Dean Ninalga, Chiranjeeve Prasad Koirala, **Sovesh Mohapatra**, Gottfried Schlaug, Marek Wodzinski, Henning Muller, David G. Ellis, Michele R. Aizenberg, M. Arda Aydın, Elvin Abidinli, Gozde Unal, Nazanin Tahmasebi, Kumaradevan Punithakumar, Tian Song, Yun Peng, Sara V. Bates, Randy Hirschtick, P. Ellen Grant, and Yangming Ou. **BONBID-HIE 2023: Lesion Segmentation Challenge in BOSTON Neonatal Brain Injury Data for Hypoxic Ischemic Encephalopathy.** *IEEE Transactions on Medical Imaging, 45(4), 1711–1725.* 2026.

Tianjia Zhu, **Sovesh Mohapatra**, Shufang Tan, Minhui Ouyang, and Hao Huang. **Mapping functional brain organization using artificial intelligence.** *Chemical & Biomedical Imaging, 4(4), 469–484.* 2026.

Minhui Ouyang, Matthew T. Whitehead, **Sovesh Mohapatra**, Tianjia Zhu, and Hao Huang. **Machine-learning based prediction of future outcome using multimodal MRI during early childhood.** *Seminars in Fetal and Neonatal Medicine*. 2024.

Anant Shinde[†], **Sovesh Mohapatra**[†], and Gottfried Schlaug. **Identifying the engagement of a brain network during a targeted tDCS-fMRI experiment using a machine learning approach.** *PLOS Computational Biology*. 2023.

Chiranjeev Prasad Koirala[†], **Sovesh Mohapatra**[†], and Gottfried Schlaug. **An ensemble approach for segmentation of neonatal HIE lesions.** *AI for Brain Lesion Detection and Trauma Video Action Recognition (BONBID-HIE 2023)*, LNCS 14567, pp. 23–27. Springer, 2025.

Sovesh Mohapatra, Advait Gosai, and Gottfried Schlaug. **SAM vs BET: A comparative study for brain extraction and segmentation of magnetic resonance images using deep learning.** *Medical Imaging meets NeurIPS*. 2023.

Chiranjeev Prasad Koirala[†], **Sovesh Mohapatra**[†], Advait Gosai, and Gottfried Schlaug. **Automated ensemble-based segmentation of adult brain tumors: A novel approach using the BraTS Africa Challenge data.** *arXiv preprint*. 2023.

Shashidhar Reddy Javaji[†], Advait Gosai[†], **Sovesh Mohapatra**[†], and Gottfried Schlaug. **Automated ensemble method for pediatric brain tumor segmentation.** *Brain Tumor Segmentation, and Cross-Modality Domain Adaptation for Medical Image Segmentation (BraTS 2023)*, LNCS 14669, pp. 211–220. Springer, 2024.

Sovesh Mohapatra[†] and Somesh Mohapatra[†]. **Sentiment is all you need to win US presidential elections.** *NLP4DH, ACL-IJCNLP*. 2022.

Sovesh Mohapatra[†] and Somesh Mohapatra[†]. **The (In)Effectiveness of intermediate task training for domain adaptation and cross-lingual transfer learning.** *arXiv preprint*. 2022.

Sovesh Mohapatra, Prathul Nath, Manisha Chatterjee, Neeladrisingha Das, Deepjyoti Kalita, Partha Roy, and Soumitra Satapathi. **Repurposing therapeutics for COVID-19: Rapid prediction of commercially available drugs through machine learning and docking.** *PLOS ONE*. 2020.

In Preparations

Sovesh Mohapatra, Steven Meisler, Fabio Pasqualetti, Theodore D. Satterthwaite, and Dani S. Bassett. **Brain Network Control Across the Human Lifespan.** *In preparation*. 2026.

Xiaohuan Xia[†], **Sovesh Mohapatra**[†], David Lydon-Staley, and Dani S. Bassett. **Citation homophily in science, before and after large language models.** *In preparation*. 2026.

Sovesh Mohapatra, David Lydon-Staley, and Dani S. Bassett. **What changes between preprint and the journal? Quantifying the effect of peer review.** *In preparation*. 2026.

[†]Contributed equally as co-first author.

RESEARCH EXPERIENCE

2025 – Present **Graduate Research Assistant**, *Complex Systems Lab, University of Pennsylvania*
With Prof. Dani S. Bassett

- Developed QUIET, open-source edge-centric network-control software for energy-efficient synchronization across synthetic and empirical data.
- Analyzed **~21 million** scientific texts with NLP to quantify militarized language and its relationship to downstream attention.
- Lead ongoing projects on Alzheimer's network energetics, citation homophily, and peer-review-driven changes from preprints to journals.

- 2023 – 2025 **Graduate Research Assistant**, *LINBC Lab, University of Pennsylvania*
 With Prof. Hao Huang
- Developed TReND, a self-supervised transformer-autoencoder and RNMF framework for neonatal functional network delineation using **300+** dHCP subjects.
 - Modeled 2-year neurodevelopmental outcomes in mild HIE from neonatal white-matter microstructure.
- 2021 – 2023 **Research Assistant**, *Schlaug Lab, University of Massachusetts Amherst*
 With Prof. Gottfried Schlaug
- Analyzed tDCS-fMRI in **33** participants to classify arcuate fasciculus network engagement across stimulation montages (best reported accuracy **0.92**).
 - Developed transfer-learning ensembles for stroke-lesion segmentation, evaluated on **22** independent T1w MRI scans (best Dice **0.736**), and engineered **lesseg** software.
 - Benchmarked SAM against BET for MRI brain extraction across diverse sequences, lesion profiles, and image-quality conditions.
- 2021 **Research Intern**, *Massachusetts Institute of Technology, Computer Science & Artificial Intelligence Laboratory*
 With Prof. Arvind Satyanarayan
- Built a deep-learning model generating natural-language descriptors from data visualizations.
 - Developed sentiment-classification models for U.S. presidential-election discourse.
- 2020 **Research Assistant**, *Satapathi Lab, Indian Institute of Technology Roorkee*
 With Prof. Soumitra Satapathi
- Built Naive Bayes and docking workflows trained on **290,893** PubChem compounds to screen **2,388** FDA-approved drugs for COVID-19 repurposing.

INVITED TALKS & PRESENTATIONS

- 2026 Invited seminar, *Yale MRRC Neuroscience Seminar Series* (Fall 2026).
QUIET: Quantifying Underutilized Influential Edges for Targeted synchronization
- 2026 Oral presentation, *Network Neuroscience Satellite Event at Network Science*.
TReND: Transformer-derived features & Regularized NMF for Functional Network Delineation
- 2025 Oral presentation, *Organization for Human Brain Mapping (OHBM)*.
- 2025 Poster, *ISMRM & ISMRT Annual Meeting*.
Predicting 2-year neurodevelopmental outcomes in mild HIE from neonatal white-matter microstructure
- 2025 Oral presentation, *ISMRM & ISMRT Annual Meeting*.
GRAFP: Graph-kernel assisted robust functional brain parcellation
- 2024 Power pitch, *ISMRM & ISMRT Annual Meeting*.
Identifying the engagement of a brain network during a targeted tDCS-fMRI experiment using a machine learning approach
- 2022 Oral presentation, *UMass Interdisciplinary Neuroscience Conference*.

TEACHING

Fall 2026	Guest Lecturer , <i>BE 5660: Networked Neuroscience</i>
Spring 2026	Teaching Assistant , <i>STAT 4850/5850: Foundations of AI: Deep Learning with Applications</i> With Prof. Enric Boix
Spring 2026	Guest Lecturer , <i>STAT 4850/5850: Foundations of AI: Deep Learning with Applications</i>

MENTORING

2026 – Present	Isabella Llano Aristizabal , master's student, <i>University of Pennsylvania</i> Research mentoring in Alzheimer's network energetics.
2022 – 2023	Advait Gosai , undergraduate student, <i>University of Massachusetts Amherst</i> Research mentoring in brain extraction and lesion segmentation; current: Research Engineer, Scale AI.
2022 – 2023	Aleksei Rutkovskii , undergraduate student, <i>University of Massachusetts Amherst</i> Research mentoring in transfer learning for brain-lesion segmentation; current: master's student, University of Massachusetts Amherst.
2022 – 2023	Chiranjeevee Prasad Koirala , master's student, <i>University of Massachusetts Amherst</i> Research mentoring in neonatal and adult brain-lesion segmentation; current: Software Engineer III, CrowdStrike.
2022 – 2023	Shashidhar Reddy Javaji , master's student, <i>University of Massachusetts Amherst</i> Research mentoring in pediatric brain-tumor segmentation; current: Ph.D. student, Stevens Institute of Technology.

AWARDS & HONORS

2026	Rising Stars in Engineering in Health.
2025	Merit Abstract Award, <i>OHBM 2025</i> .
2025	Distinguished Alumni, SIP Abacus.
2025	Trainee Award, <i>ISMRM & ISMRT Annual Meeting</i> .
2024	Trainee Award, <i>ISMRM & ISMRT Annual Meeting</i> .
2022	Best Poster Award, <i>UMass Interdisciplinary Neuroscience Conference</i> .
2021	UMass Core Summer Fellow.
2019	YonoSBI 20 under 20, Arts & Literature.
2016	President of India Medal for Exceptional Performance in Innovation.

LEADERSHIP & SERVICE

2026	Guest Editor , special issue on Pediatric MRI, <i>Magnetic Resonance in Medicine</i> .
2025 – 2026	Trainee Representative , <i>ISMRM Pediatric Study Group</i> . Represent trainee interests; facilitate communication between trainees and senior researchers.
2023 – 2025	Co-Chair / Co-President , <i>Gordon Research Seminar — Tissue Microstructure Imaging</i> . Organized a conference for graduate and postdoctoral researchers spanning multiple tissue-microstructure imaging modalities.
2023 – 2024	Advocacy Chair , <i>Penn Rangoli</i> . Supported advocacy and community programming for South Asian students at Penn; helped organize cultural, educational, and community-building events.

2021 – 2023	Resident Advisor , <i>Brett Hall, University of Massachusetts Amherst</i> . Fostered an inclusive residential community, supported student well-being, and coordinated programming for undergraduate residents.
2021 – 2023	Member , <i>Anti-Racism/Bias Training Committee, University of Massachusetts Amherst</i> . Prepared a five-hour training module for incoming resident assistants and peer mentors on recognizing incidents of racism and bias and responding effectively.
2022 – 2023	Member , <i>Dean’s Student Advisory Council, College of Natural Sciences, University of Massachusetts Amherst</i> . Represented student perspectives to college leadership and contributed feedback on academic programs, student support, and campus climate initiatives.
Reviewing	MICCAI (2024–2026); OHBM (2025); ISMRM (2024–2025); <i>Human Brain Mapping</i> ; <i>Brain Research</i> ; Gordon Research Conference; ML4LMS, ICML (2024).

OUTREACH & PUBLIC ENGAGEMENT

2019	LinkedIn Local, GLA University.
2018, 2019	TEDx Speaker — TEDxNITSrinagar, TEDxRamjasCollege, TEDxManipal.

REFERENCES

Dani S. Bassett

Wu Tsai Professor

Bioengineering

Yale University[send.Bassett.60C4B7C5D2@](mailto:send.Bassett.60C4B7C5D2@interfoliodossier.com)[interfoliodossier.com](mailto:send.Bassett.60C4B7C5D2@interfoliodossier.com)**David M. Lydon-Staley**

Associate Professor

Annenberg School for Communica-
tion*University of Pennsylvania*[send.LydonStaley.6A6262ECA4@](mailto:send.LydonStaley.6A6262ECA4@interfoliodossier.com)[interfoliodossier.com](mailto:send.LydonStaley.6A6262ECA4@interfoliodossier.com)**Gottfried Schlaug**

Scientific Director

Human Magnetic Resonance Cen-
ter*University of Massachusetts**Amherst*[send.Schlaug.7798EC09A3@](mailto:send.Schlaug.7798EC09A3@interfoliodossier.com)[interfoliodossier.com](mailto:send.Schlaug.7798EC09A3@interfoliodossier.com)